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Final Project Proposal: A Comparison of Radar-Based Hail Size Estimation Algorithms

I. Literature Review

Hail is the costliest and most frequent convective hazard within the United States, inflicting billions of dollars in insured economic losses annually (Battaglioli et al. 2026). Supercell thunderstorms account for the overwhelming majority of large hail (Blair et al. 2011), reflected by a peak climatological frequency of severe and significant-severe hailfall within the Great Plains (Cintineo et al. 2012; Murillo et al. 2023). Therefore, methods of estimating hailfall are crucial for proactive hail nowcasting and risk assessment that can mitigate economic loss, particularly in Great Plains states such as Oklahoma. Point-based reports at the surface have often been used for climatological and warning verification purposes but are plagued by numerous non-meteorological biases and limitations, which are related to factors such as population density, diurnal cycles, and reference objects, that contribute to unreliability in magnitude and inhomogeneity in spatiotemporal coverage (Doswell et al. 2005; Allen and Tippett 2015; Wendt and Jirak 2021). Moreover, hail reports are retrieved retroactively and thus cannot be used for proactive warnings. Since the advent of Doppler radar, radar-based hail retrieval methods have sought to remedy these limitations via uniform objective gridded output, though systematic issues such as resonance effects from Mie scattering (particularly for large hail and lower-wavelength radar bands), attenuation (particularly for lower-wavelength radars), and the overlapping distributions of hydrometeors have limited their utility (Homeyer et al. 2023).

Due to their inability to characterize the shape of hydrometeors beyond one dimension, initial single-polarization radar methods struggled to detect and quantify hail size beyond simple relationships with reflectivity factor (Z_H), which highly depends on scatterer diameter (by a factor of six) yet is prone to resonance effects. Among the most prominent hail retrieval fields is the maximum estimated size of hail (MESH), which is derived from a power-law relationship between thermally-weighted, vertically-integrated reflectivity above the melting layer and a limited set of hail reports (147) in Oklahoma and Florida (Witt et al. 1998). While literature has demonstrated the utility of MESH for spatial hail assessment and general detection, particularly at night and in areas with low population density when human-initiated reports are minimized (Wendt and Jirak 2021), MESH was intentionally fit to be an overestimate of maximum hail size (75th percentile) and is overall a poor predictor of maximum hail size (Ortega 2018). While Murillo and Homeyer (2019) modified the coefficients of the power-law relationship for a significantly larger (5,954) and more representative set of hail reports within the U.S., limitations of MESH as a predictor of hail size remain given its Z_H dependence.

The introduction of dual polarization has enabled radar fields that can characterize hydrometeor shape and diversity, including differential reflectivity (Z_{DR}), which is the difference between reflectivity at horizontal and vertical polarizations. Since hailstones generally exhibit different shapes, and thus polarimetric characteristics, relative to rain drops (i.e. more symmetrical as opposed to oblate), including Z_{DR} closer to zero and reduced cross-correlation coefficient (ρ_{HV}), these polarimetric differences contribute to hail retrieval methods such as hail differential reflectivity (H_{DR}) and the hail size discrimination algorithm (HSDA).

H_{DR} is a simple relationship between Z_H and Z_{DR} derived using raindrop size distributions from disdrometer measurements to distinguish hail from rain (Aydin et al. 1986), which has demonstrated greater skill for detecting severe hail relative to single-polarization methods (Depue et al. 2007; Murillo and Homeyer 2019; Brook et al. 2021). While HDR does not directly estimate hail size, Depue et al. (2007) applied a least-squares fit of HDR to hail diameter, the equation for which was approximated by Brook et al. (2021). HSDA, meanwhile, is an extension of the hydrometeor classification algorithm (HCA) from Park et al. (2009), with three categorical hail size bins: small hail (diameter < 25 mm), large hail (25 mm < diameter < 50 mm), and giant hail (>50 mm) (Ortega et al. 2016). The HCA uses a fuzzy-logic scheme, which accounts for overlapping radar field distributions among hydrometeor types via weighted membership functions, to determine the dominant hydrometeor type within a volume (Park et al. 2009). Unlike other HCA methods, Park et al. (2009) also weights the membership functions by a confidence vector **Q**, which accounts for measurement errors due to factors such as attenuation and nonuniform beam filling. For volumes classified as “a mixture of rain and hail (RH)” by the HCA, the HSDA determines the hail size bin using a similar scheme, where weighted membership functions for distributions of Z_H, Z_{DR}, and ρ_{HV} at various thermal height intervals are used (Ortega et al. 2016).

While radar-based hail retrievals represent hail lofted within a thunderstorm, hailfall at the surface is often displaced due to horizontal advection by low-level winds and melting as hailstones fall below the freezing level. With advances in three-dimensional wind retrieval via dual-Doppler analysis (DDA), radar-derived hailstone trajectories can account for the spatial offset often observed between radar-based hail proxies and surface reports arising from horizontal advection from low-level winds during hailfall (Brook et al. 2021). Moreover, hail trajectories may enable the collection of novel statistics such as impact angle (difference between incident hailstone angle and the vertical axis at impact), which may be representative of the type of hail damage received (e.g. siding for larger impact angles, and vice versa for roofing) and thus relevant to end users such as insurers (Brook et al. 2021). Numerical simulations of hailstorms frequently utilize hailstone trajectory models (Kumjian and Lombardo 2020), yet fewer utilize trajectories for real-world cases (Pounds et al. 2024), particularly in operational settings outside of targeted field campaigns (Brook et al. 2021). Using a coupled hailstone melting model and DDA-based trajectory model, named HailTrack, for a significant hail event in Australia, Brook et al. (2021) reduced the spatial offset from conventional radar-based hail swaths relative to high-resolution insurance claims data, with the greatest performance for advection-corrected HDR that exceeded reflectivity, MESH, and HSDA-based methods.

However, three-dimensional wind retrieval via Doppler analysis is a non-trivial task, and the methodology can substantially affect results. While single Doppler methods do exist, assimilation methods using multiple radars and/or passive receivers are far superior (Protat et al. 2024). Among the most commonly used dual-Doppler techniques is the three dimensional variational technique (3DVAR), which obtains the wind vector field by minimizing a cost function J(**V**), which itself is a weighted sum of cost functions associated with various physical constraints, such as the mass continuity and vertical vorticity equations, and alternative sources, such as model or radiosonde wind fields (Potvin et al. 2012; Jackson et al. 2020). Even the decision to map radar fields onto a common Cartesian grid via Cressman or Barnes spatial interpolation, as often required for methods such as the 3DVAR technique within the open-source Python package PyDDA, can significantly degrade wind retrievals and associated features relative to direct assimilation (Brook et al. 2023). Moreover, even multi-Doppler techniques such as 3DVAR struggle to represent the true wind field, particularly for deep convection (Protat et al. 2024), which is directly pertinent to hail production.

II. Project Goals

Given the array of radar-based hail retrieval algorithms of varying complexity, the primary goal of this study is to quantitatively and qualitatively compare the spatial extent and maximum intensity of hail estimated by the radar algorithms relative to report-based ground reports. This study will primarily evaluate the maximum estimated size of hail (MESH; Witt et al. 1998), hail differential reflectivity (H_{DR}; Aydin et al. 1986), and the hail size discrimination algorithm (HSDA; Ortega et al. 2016) for a variety of cases with available polarimetric S-band radars and extensive ground-based reports for verification.

Time permitting, the trajectory-based radar hail algorithm demonstrated in Brook et al. (2021) will be partially implemented to examine whether accounting for the advection of lofted hailstones will improve the spatial distribution of hailfall. Given the limited timeframe of this project, the lack of existing open-source Python resources for HailTrack and radar-based microphysical models, and the complexity surrounding the appropriate implementation of microphysical schemes, the melting model will be excluded from this analysis. Through sensitivity tests, however, Brook et al. (2021) demonstrate minimal degradation to overall performance when melting is neglected in the algorithm.

Therefore, this study seeks to address the following research questions:

- Which radar-based hail retrieval algorithm (HSDA, H_{DR}, or MESH) most accurately represents the spatial extent and relative intensity of hail for hail-producing supercells in central Oklahoma?

- Do algorithms utilizing polarimetric variables (HSDA, H_{DR}) provide substantial value in estimating hail size relative to single-polarization methods (MESH)?
- Time permitting, can a rudimentary advection-correcting trajectory-based algorithm improve the spatial distribution of radar-estimated hailfall approaches despite the limitations of dual-Doppler retrieval of three-dimensional winds?

Given the results of Brook et al. (2021), along with the utility of dual-polarization products for hail retrieval demonstrated in other studies (Murillo and Homeyer 2019; Homeyer et al. 2023), it is hypothesized that H_{DR} would most accurately represent relative hail intensity bins, followed by HSDA and MESH respectively, due to Z_{DR}'s ability to capture multi-dimensional hydrometeor shape and Z_H's strong dependence on hydrometeor diameter. However, performance will likely be degraded owing to the inherent flaws of Mie scattering and an imperfect validation dataset. Additionally, the trajectory-based algorithm utilizing H_{DR} for initial hail size is hypothesized to better represent the spatial distribution of surface hail reports given its ability to correct for advection by the three-dimensional wind field, though such results are likely sensitive to the chosen grid spacing (higher HSS for smaller grid spacing).

III. Proposed Methods

Considering its high spatial density (median spacing of 1.74 km), maximum surface hail reports from the Severe Hazards Analysis & Verification Experiment (SHAVE; Ortega et al. 2009; Ortega 2018) will be combined with *Storm Data* reports as robust ground-truth verification of the hail algorithms. With the requirement of dual-polarization capabilities for H_{DR} and HSDA (post 2012), the temporal domain of SHAVE (through 2015), the proximity of multiple S-band radars (KTLX, KOUN, and KCRI) for reliable dual-Doppler retrievals, and the desire to examine a broad range of hail sizes, the largest of which arise almost exclusively from supercells (Blair et al. 2011), the search domain for this study is limited from 2013 to 2015 in the Oklahoma City metro area (Canadian, Cleveland, McClain, and Oklahoma counties). Specifically, the following cases are tentatively chosen:

- 29 March 2013
  - Hail diameter range: 0-2" (13 SHAVE reports, 13 local storm reports [LSRs])
  - Counties: Canadian
  - Radars: KTLX, KCRI (i.e. FOP1)
- 19 May 2013
  - Hail diameter range: 0-3" (29 SHAVE reports, 19 LSRs)
  - Counties: McClain, Cleveland, Oklahoma
  - Radars: KTLX, KCRI (i.e. NOP4)
- 30 May 2013
  - Hail diameter range: 0-1" (30 SHAVE reports, 4 LSRs)
  - Counties: McClain, Cleveland
  - Radars: KTLX, KCRI (i.e. DAN1)

Given the limited quantity of surface reports (108) and cases (3) for validation, which may reduce confidence in the results, the following dates with a sufficiently broad spatial and size distribution of non-SHAVE storm reports are also considered for analysis: 23 March 2019 (16 LSRs), 4 May 2020 (21 LSRs), 31 August 2020 (21 LSRs), 28 April 2021 (16 LSRs), 10 October 2021 (10 LSRs), 19 April 2023 (46 LSRs), 24 September 2024 (37 LSRs), 28 April 2025 (15 LSRs), 25 May 2025 (18 LSRs), and 10 March 2026 (43 LSRs).

Level II polarimetric radar data from KTLX and KCRI for the temporal domain (2013–present) are accessible via the NEXRAD AWS archive (registry.opendata.aws/noaa-nexrad/), with KCRI listed under numerous configuration identifiers (i.e. NOP3, NOP4, ROP3, ROP4, FOP1, DAN1, and DOP1) until December 2025 (Clark 2025). SHAVE data are stored locally by the author, and quality-controlled *Storm Data* reports are available via NCEI (ncei.noaa.gov/stormevents/ftp.jsp). All radar output and hail reports will be mapped onto a common Cartesian grid via Py-ART in order to directly compare performance across multiple radar sites and coordinate systems (point-based versus polar). Specifically, polar radar data will be gridded according to the two-pass Barnes interpolation scheme (Barnes 1964; Majcen et al. 2008) due to its predictable behavior and suitability for multi-Doppler analyses (Majcen et al. 2008), with a common horizontal resolution of 1 km and vertical resolution of 500 m.

While Py-ART contains a built-in function for semi-supervised hydrometeor classification (HC) from Besic et al. (2016), it does not natively support the complementary HSDA, nor either the MESH or HDR fields. However, the pyhail package enables the trivial retrieval of these fields within the Py-ART framework (Solderholm 2026). The HSDA will be applied for the "melting hail" class from Besic et al. (2016), as it is most similar to the mixed phase "RH" class from Park et al. (2009) used in Ortega et al. (2016). Due to HSDA's dependence on HC, which necessitates a vertical thermodynamic profile for Py-ART's Besic et al. (2016) implementation, tabular radiosonde data will be obtained from the Iowa Environmental Mesonet's Rawinsonde Data Archive for KOUN (Norman, OK) at the nearest initialization time (commonly 12Z or 00Z) prior to each event (mesonet.agron.iastate.edu/archive/raob/). Moreover, the confidence vector **Q** used in Park et al. (2009) and Ortega et al. (2016) will be neglected due to its absence in pyhail and the Besic et al. (2016) HC algorithm. Since Py-ART's HC algorithm is also a function of specific differential phase (K_{DP}), the Vulpiani method (Vulpiani et al. 2012) will be applied to obtain the K_{DP} field from Φ_{DP}, with the optimal configuration (windsize=10, n_iter=2) from Reimel and Kumjian (2021).

Since H_{DR} is measured in decibels (rather than a direct measure of hail size), prior research has demonstrated MESH's imprecision in estimating hail size (Ortega 2018), and the distribution of numerical *Storm Data* hail reports is effectively discretized (Doswell et al. 2005), radar algorithms will be evaluated according to verification statistics, namely Heidke skill score (HSS) due to its ubiquity in literature (Brook et al. 2021), that are derived from binary contingency tables. To produce binary (0/1) bins for the continuous output of MESH and H_{DR}, the optimal value thresholds for each hail algorithm, defined by maximum HSS, will be determined for the fixed thresholds of the validation data (storm reports), as common in literature (Cintineo et al. 2012; Brook et al. 2021; Murillo et al. 2023). In considering maximum hail intensity, performance will be evaluated for two pre-defined categories: severe hail (≥1 in., or 25.4 mm, in diameter) and significant-severe (≥2 in., or 50.8 mm), which are standard thresholds for U.S. hail studies (Cintineo et al. 2012; Murillo et al. 2023). Although the discrete categorical output of HSDA differs from the continuous numerical output of H_{DR} and MESH, the bounds for the "large" and "giant" hail categories (25 mm and 50 mm, respectively) are within one millimeter of the severe and significant-severe thresholds. Thus, the "large" and "giant" HSDA categories will be considered severe, and "giant" will be considered significant-severe.

Time permitting, a rudimentary algorithm partially resembling the HailTrack algorithm (Brook et al. 2021), excluding the melting scheme, will be applied. Following Brook et al. (2021), this will involve a multi-step process, beginning with three-dimensional wind retrievals via dual-Doppler analysis from the gridded radar data, estimation of initial lofted maximum hailstone location and sizes, and finally the trajectory analysis using the three-dimensional wind field and initial hailstone locations. The Pythonic Direct Data Assimilation (PyDDA) package (Jackson et al. 2020), which implements the variational data assimilation techniques from Potvin et al. (2012) and Shapiro et al. (2009), and is compatible with Py-ART, will be used to retrieve the three-dimensional wind field for each scan. The lofted "hailstones" will be initialized once per pixel via H_{DR}. Finally, a system of three ordinary differential equations (ODEs) will be solved for each initial hailstone position for a given timestep, with one equation for each axis, where the only forces acting on the hailstone are gravity and drag in the vertical. Horizontal position (x, y) is governed solely by horizontal advection by the u or v wind, respectively, while vertical position is determined from advection by the sum of ambient vertical velocity (w) and terminal velocity of the hailstone (V_T), as determined by gravity and drag (Brook et al. 2021).

References

Allen, J. T., and M. K. Tippett, 2015: The Characteristics of United States Hail Reports: 1955-2014. *EJSSM*, **10**, 1–31, https://doi.org/10.55599/ejssm.v10i3.60.

Aydin, K., T. A. Seliga, and V. Balaji, 1986: Remote Sensing of Hail with a Dual Linear Polarization Radar. *J. Climate Appl. Meteor.*, **25**, 1475–1484, https://doi.org/10.1175/1520-0450(1986)025<1475:RSOHWA>2.0.CO;2.

Battaglioli, F., M. Taszarek, P. Groenemeijer, T. Púčik, and A. Rädler, 2026: Contrasting trends in very large hail events and related economic losses across the globe. *Nat. Geosci.*, **19**, 52–58, https://doi.org/10.1038/s41561-025-01868-0.

Besic, N., J. Figueras I Ventura, J. Grazioli, M. Gabella, U. Germann, and A. Berne, 2016: Hydrometeor classification through statistical clustering of polarimetric radar measurements: a semi-supervised approach. *Atmos. Meas. Tech.*, **9**, 4425–4445, https://doi.org/10.5194/amt-9-4425-2016.

Blair, S. F., D. R. Deroche, J. M. Boustead, J. W. Leighton, B. L. Barjenbruch, and W. P. Gargan, 2021: Radar-Based Assessment of the Detectability of Giant Hail. *EJSSM*, **6**, 1–30, https://doi.org/10.55599/ejssm.v6i7.34.

Brook, J. P., A. Protat, J. Soderholm, J. T. Carlin, H. McGowan, and R. A. Warren, 2021: HailTrack—Improving Radar-Based Hailfall Estimates by Modeling Hail Trajectories. *Journal of Applied Meteorology and Climatology*, **60**, 237–254, https://doi.org/10.1175/JAMC-D-20-0087.1.

Brook, J. P., A. Protat, C. K. Potvin, J. S. Soderholm, and H. McGowan, 2023: The Effects of Spatial Interpolation on a Novel, Dual-Doppler 3D Wind Retrieval Technique. *Journal of Atmospheric and Oceanic Technology*, **40**, 1325–1347, https://doi.org/10.1175/JTECH-D-23-0004.1.

Cintineo, J. L., T. M. Smith, V. Lakshmanan, H. E. Brooks, and K. L. Ortega, 2012: An Objective High-Resolution Hail Climatology of the Contiguous United States. *Weather and Forecasting*, **27**, 1235–1248, https://doi.org/10.1175/WAF-D-11-00151.1.

Clark, T. J., 2025: Updated: Radar Operations Center Testbed Configuration Identifier Changes: Effective on or about September 29, 2025, https://www.weather.gov/media/notification/pdf_2025/scn25-67_Updated_KCRI_Configuration_Name_Changes_aaa.pdf.

Depue, T. K., P. C. Kennedy, and S. A. Rutledge, 2007: Performance of the Hail Differential Reflectivity (HDR) Polarimetric Radar Hail Indicator. *Journal of Applied Meteorology and Climatology*, **46**, 1290–1301, https://doi.org/10.1175/JAM2529.1.

Doswell, C. A., H. E. Brooks, and M. P. Kay, 2005: Climatological Estimates of Daily Local Nontornadic Severe Thunderstorm Probability for the United States. *Weather and Forecasting*, **20**, 577–595, https://doi.org/10.1175/WAF866.1.

Homeyer, C. R., E. M. Murillo, and M. R. Kumjian, 2023: Relationships between 10 Years of Radar-Observed Supercell Characteristics and Hail Potential. *Monthly Weather Review*, **151**, 2609–2632, https://doi.org/10.1175/MWR-D-23-0019.1.

Jackson, R., S. Collis, T. Lang, C. Potvin, and T. Munson, 2020: PyDDA: A Pythonic Direct Data Assimilation Framework for Wind Retrievals. *JORS*, **8**, 20, https://doi.org/10.5334/jors.264.

Kumjian, M. R., and K. Lombardo, 2020: A Hail Growth Trajectory Model for Exploring the Environmental Controls on Hail Size: Model Physics and Idealized Tests. *Journal of the Atmospheric Sciences*, **77**, 2765–2791, https://doi.org/10.1175/JAS-D-20-0016.1.

Murillo, E. M., and C. R. Homeyer, 2019: Severe Hail Fall and Hailstorm Detection Using Remote Sensing Observations. *Journal of Applied Meteorology and Climatology*, **58**, 947–970, https://doi.org/10.1175/JAMC-D-18-0247.1.

Murillo, E. M., C. R. Homeyer, and J. T. Allen, 2021: A 23-Year Severe Hail Climatology Using GridRad MESH Observations. *Monthly Weather Review*, **149**, 945–958, https://doi.org/10.1175/MWR-D-20-0178.1.

Ortega, K. L., T. M. Smith, K. L. Manross, K. A. Scharfenberg, A. Witt, A. G. Kolodziej, and J. J. Gourley, 2009: THE SEVERE HAZARDS ANALYSIS AND VERIFICATION EXPERIMENT. *Bull. Amer. Meteor. Soc.*, **90**, 1519–1530, https://doi.org/10.1175/2009BAMS2815.1.

Ortega, K. L., J. M. Krause, and A. V. Ryzhkov, 2016: Polarimetric Radar Characteristics of Melting Hail. Part III: Validation of the Algorithm for Hail Size Discrimination. *Journal of Applied Meteorology and Climatology*, **55**, 829–848, https://doi.org/10.1175/JAMC-D-15-0203.1.

Ortega, K. L., 2018: Evaluating Multi-Radar, Multi-Sensor Products for Surface Hailfall Diagnosis. *EJSSM*, **13**, 1–36, https://doi.org/10.55599/ejssm.v13i1.69.

Park, H. S., A. V. Ryzhkov, D. S. Zrnić, and K.-E. Kim, 2009: The Hydrometeor Classification Algorithm for the Polarimetric WSR-88D: Description and Application to an MCS. *Weather and Forecasting*, **24**, 730–748, https://doi.org/10.1175/2008WAF2222205.1.

Potvin, C. K., A. Shapiro, and M. Xue, 2012: Impact of a Vertical Vorticity Constraint in Variational Dual-Doppler Wind Analysis: Tests with Real and Simulated Supercell Data. *Journal of Atmospheric and Oceanic Technology*, **29**, 32–49, https://doi.org/10.1175/JTECH-D-11-00019.1.

Pounds, L. E., C. L. Ziegler, R. D. Adams-Selin, and M. I. Biggerstaff, 2024: Analysis of Hail Production via Simulated Hailstone Trajectories in the 29 May 2012 Kingfisher, Oklahoma, Supercell. *Monthly Weather Review*, **152**, 245–276, https://doi.org/10.1175/MWR-D-23-0073.1.

Protat, A., V. Louf, and J. P. Brook, 2024: SWIRL: The First Australian Operational Radar-Based 3D Wind Analysis System. *Journal of Atmospheric and Oceanic Technology*, **41**, 725–746, https://doi.org/10.1175/JTECH-D-23-0155.1.

Solderholm, Joshua, 2026: Python Hail Retrieval Toolkit (Version 3.3.2). https://github.com/joshua-wx/pyhail/releases/tag/v3.3.2.

Vulpiani, G., M. Montopoli, L. D. Passeri, A. G. Gioia, P. Giordano, and F. S. Marzano, 2012: On the Use of Dual-Polarized C-Band Radar for Operational Rainfall Retrieval in Mountainous Areas. *Journal of Applied Meteorology and Climatology*, **51**, 405–425, https://doi.org/10.1175/JAMC-D-10-05024.1.

Wendt, N. A., and I. L. Jirak, 2021: An Hourly Climatology of Operational MRMS MESH-Diagnosed Severe and Significant Hail with Comparisons to Storm Data Hail Reports. *Weather and Forecasting*, **36**, 645–659, https://doi.org/10.1175/WAF-D-20-0158.1.

Witt, A., M. D. Eilts, G. J. Stumpf, J. T. Johnson, E. D. W. Mitchell, and K. W. Thomas, 1998: An Enhanced Hail Detection Algorithm for the WSR-88D. *Wea. Forecasting*, **13**, 286–303, https://doi.org/10.1175/1520-0434(1998)013<0286:AEHDAF>2.0.CO;2.